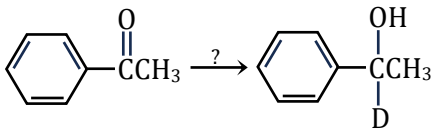
RACE Regular Analysis Through Continuous Exercise

Reduction : RACE-01

Metal hydride reduction ($LiAlH_4$; $NaBH_4$; DIBAL-H; B_2H_6; $LiBH_4$; $NaBH_3CN$)

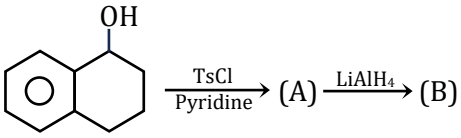
1. Which of the following reagents would carry out the following transformation ? (D = 2H)



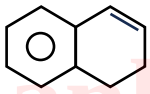
(A) $NaBD_4$ in CH_3OH (B) $LiAlH_4$, then D_2O

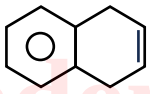
(C) $NaBD_4$ in CH_3OD (D) $LiAlD_4$, then D_2O

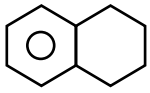
CRD120

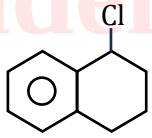
2. 

Product (*B*) of the above reaction is :

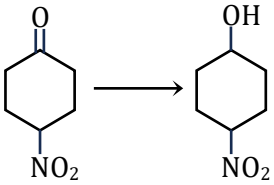
(A) 

(B) 

(C) 

(D) 

CRD121

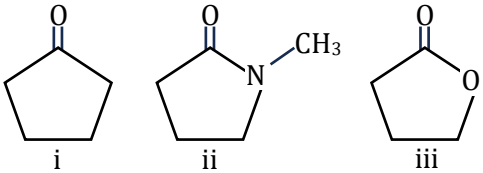
3. 

Above conversion can be achieved by :

(A) $LiAlH_4$ (B) $NaBH_4$ (C) H_2/Ni (D) CrO_3

CRD122

4. Arrange the following compounds in order of their reactivity toward $LiAlH_4$.

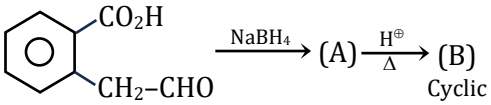


(A) i < ii < iii (B) i < iii < ii (C) ii < i < iii (D) ii < iii < i

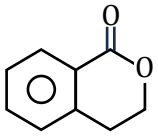
CRD123

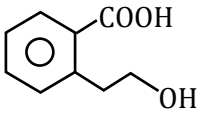
Rankers Academy JEE

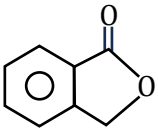
For More Join: @IITJEE_Advanced

5. 

Compound (*B*) is :

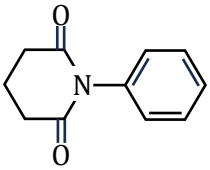
(A) 

(B) 

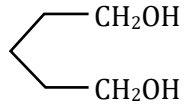
(C) 

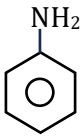
(D) (A) and (C) both

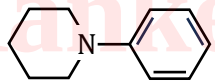
CRD124

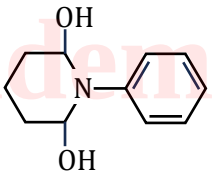
6. 

$\xrightarrow[H_2O]{LiAlH_4(excess)}$ Product of the reaction is :

(A) 

(B) 

(C) 

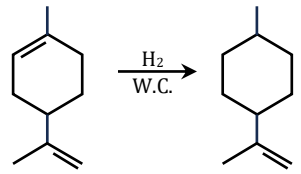
(D) 

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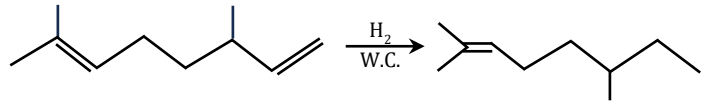
CRD128

Catalytic hydrogenation (Except alkene and alkyne)

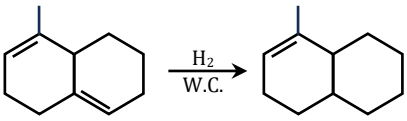
7. In the given selective hydrogenation which combination is incorrect ?

(A) 

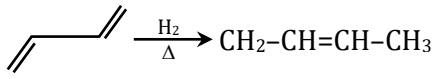
(W.C. = Wilkinsons catalyst)

(B) 

(W.C. = Wilkinsons catalyst)

(C) 

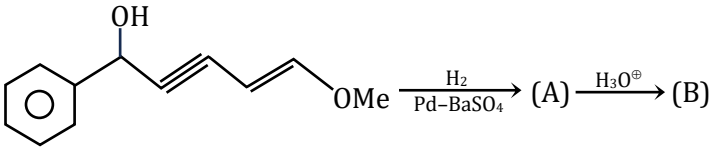
(W.C. = Wilkinsons catalyst)

(D) 

(W.C. = Wilkinsons catalyst)

CRD118

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8. 

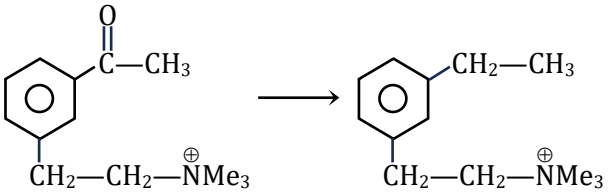
Product (B) is :

(A) Ph–CH(OH)–CH=CH–CH_2–C(=O)–H (B) Ph–CH=CH–CHO

(C) Ph–$(CH=CH)_2$–CHO (D) Ph–$(CH=CH)_3$–CHO

CRD119

Bouvealt Blanc reduction

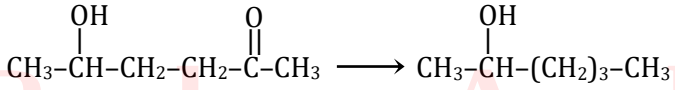
9. 

Above conversion can be achieved by :

(A) Wolf-Kishner reduction (B) Clemmensen reduction

(C) $LiAlH_4$ (D) $NaBH_4$

CRD125

10. 

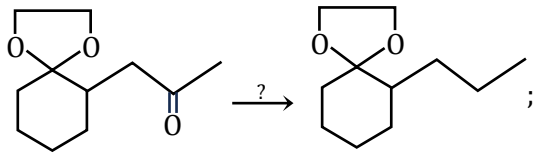
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Above conversion can be achieved by :

(A) Wolff-Kishner reduction (B) Clemmensen reduction

(C) $LiAlH_4$ (D) $NaBH_4$

CRD126

11. 

Above conversion can be carried out by :

(A) Clemmensen reduction (B) Wolff-Kishner reduction

(C) $LiAlH_4$ (D) $NaBH_4$

CRD127

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	D	A	C	A	C	B	A
Question	11									
Answer	B									

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